

Assignment 1

1 Requirements

You are expected to complete all required exercises and encouraged to complete the optional ones. For submission, please put all your answers in a PDF file. The name of the file should follow the format "studentID assignmentNumber" (e.g., 2024111054_assignment1). **The sub-mission deadline is 20241007.** Late submissions are still allowed within one week after the deadline (grace period). If you submit your assignment during the grace period, your score will be half of the score you could get if the submission was made before the deadline. Assignment submitted after the grace period will not be graded, meaning that you will get a zero for the assignment.

2 Required Exercises (100 points)

Exercise 1

When a C compiler compiles the following statement, how many tokens will it generate? [5points]

```
int a3 = a * 3;
```

Exercise 2

In a string of length n ($n > 0$), how many of the following are there?

1. Prefixes [5 points]
2. Proper prefixes [5 points]
3. Prefixes of length m ($0 < m \leq n$) [5 points]
4. Suffixes of length m ($0 < m \leq n$) [5 points]
5. Proper prefixes of length m ($0 < m \leq n$) [10 points]
6. Substrings [10 points]
7. Subsequences [10 points]

Exercise 3

Describe the languages denoted by the following regular expressions:

1. $((\epsilon|a)^*b^*)^*$ [5 points]
2. $(a|b)^*a(a|b)(a|b)$ [5 points]

3. $a^*ba^*ba^*ba^*$ [5 points]

Exercise 4

Write regular definitions or regular expressions for the following languages.

1. All strings representing valid telephone numbers in Shenzhen. A valid telephone number contains the country code (86), a hyphen, the area code 010, another hyphen, and eight digits where the first one cannot be zero (e.g., 86-010-62282045). [10 points]
2. All strings of a's and b's that start with a and end with b. [10 points]
3. All strings of lowercase letters that contain the five vowels in order. [10 points]

3 Optional Exercises (10 bonus points)

Exercise 1

Suppose we have a alphabet $\Sigma = a, b, c$, write regular definitions to describe all strings over Σ without repeated letters. [Hint: You may draw an NFA for the language and convert the NFA to regular definitions.]

Assignment 2

1 Requirements

You are expected to complete all required exercises and encouraged to complete the optional ones. For submission, please put all your answers in a PDF file. The name of the file should follow the format “**studentID_assignmentNumber**” (e.g., 2024111054_assignment2). **The submission deadline is 11:59 PM, October 13, 2024. Please ensure that your answers (including NFAs and DFAs) are clear enough, otherwise it will affect the evaluation. It is recommended to use drawing software to draw NFAs and DFAs.**

2 Required Exercises (100 points)

Exercise 1: Design NFAs and DFAs to recognize each of the following regular languages. Detailed steps can be omitted.

1. $L((a|b)^*b)$ [10 points]
1. $L(((\epsilon|a)^*b)^*)$ [10 points]
2. $L((a|b)^*a(a|b)(a|b))$ [10 points]
3. $L(a^*ba^*ba^*ba^*)$ [10 points]

Exercise 2: Convert the following regular expressions to NFAs using the **Thompson's Construction Algorithm** (Algorithm 3.23 in the dragon book). **Please put down the detailed steps.**

1. $((\epsilon|a)^*b)^*$ [10 points]
1. $(a|b)^*a(a|b)(a|b)$ [10 points]
1. $a^*ba^*ba^*ba^*$ [10 points]

Exercise 3: Convert the NFAs in Exercise2 to DFAs using the **Subset Construction Algorithm** (Algorithm 3.20 in the dragon book). **Please put down the detailed steps.** [30points in total; 10 points for each correct conversion]

3 Optional Exercises (20 bonus points)

Exercise 1: Minimize the number of states of the DFAs you have built for regular expressions 2 and 3 in Exercise 2 using the State-Minimization Algorithm (Algorithm 3.39 in the dragon book). **Please put down the detailed steps.** [10 points for each correct minimization process]

Assignment 3

1 Requirements

You are expected to complete all required exercises and encouraged to complete the optional ones. For submission, please put all your answers in a PDF file. The name of the file should follow the format "studentID_assignmentNumber"

(e.g., 2024111054_assignment3). **The submission deadline is 11:59 PM, October 27, 2024.**

2 Required Exercises (100 points)

Exercise 1 (Grammar Basics)

Consider the following context-free grammar G:

$$S \rightarrow SS + | SS^* | a$$

1. Is the string $a + a^*a$ in $L(G)$? [10 points]
2. Give a leftmost derivation for the string aa^*aa+^* . [10 points]
3. Give a rightmost derivation for the string aa^*aa+^* . [10 points]
4. Give a parse tree for the string aa^*aa+^* . [10 points]
5. Give an equivalent grammar without immediate left recursions. [10 points]

Exercise 2 (Top-Down Parsing)

Consider the following grammar G:

$$S \rightarrow aB$$

$$B \rightarrow S + B | \epsilon$$

1. Construct the predictive parsing table for G. Please put down the detailed steps, including the calculation of FIRST and FOLLOW sets. [25 points]
2. Is the grammar LL(1)? [5 points]
3. Can an LL(1) parser accept the input string $aaaa+++$? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts in the parsing table if any. [20 points]

3 Optional Exercises (10 bonus points)

Exercise 1

Consider the following context-free grammar:

短语 $\rightarrow$ 人 | 短语动词短语

动词 $\rightarrow$ 喜欢 | 讨厌

人 $\rightarrow$ 你 | 我 | 他

The grammar can produce sentences such as “我喜欢你”. Is the grammar ambiguous? If yes, please give one sentence and its multiple parse trees. If no, state the reason. [5 points for the yes/no answer and 15 points for the justification]

Assignment 4

1 Requirements

You are expected to complete all required exercises. For submission, please put all your answers in a PDF file. The name of the file should follow the format “**studentID assignmentNumber**” (e.g., 2024111554 assignment4). **The submission deadline is 11:59 PM, November 10, 2024. Please ensure that your answer process is detailed enough, otherwise it will affect the evaluation.**

2 Bottom-Up Parsing Exercises (100 points)

Exercise 1 (Simple LR): Consider the following grammar G:

$$S \rightarrow aB$$

$$B \rightarrow S + B \mid \epsilon$$

1. Construct the SLR(1) parsing table for G. Please put down the detailed steps, including the calculation of LR(0) item sets. [25 points]
2. Is the grammar SLR(1)? [10 points]
3. Can the SLR(1) parser accept the input string `aaaa+++`? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts if any. [20 points]

Exercise 2 (Canonical LR): Convert Consider the grammar G in Exercise 1:

1. Construct the CLR(1) parsing table for G. Please put down the detailed steps, including the calculation of LR(1) item sets. [25 points]
2. Can the CLR(1) parser accept the input string `aaaa+++`? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts if any. [20 points]

3 Optional Exercises (30 bonus points)

Exercise 3 (Lookahead LR): Consider the grammar G in Exercise 1:

1. Construct the LALR(1) parsing table for G. Please put down the detailed steps, including the merging of LR(1) item sets. [20 points]
2. Can the LALR(1) parser accept the input string `aaaa+++`? If yes, please list the moves made

by the parser; otherwise, state the reason. Before parsing, please resolve conflicts if any. [10 points]

Assignment 5

1 Requirements

You are expected to complete all exercises. For submission, please put all your answers in a PDF file and submit it. The name of the file should follow the format "student-ID_assignmentNumber" (e.g., 2024111054_assignment5). **The submission deadline is 11:59 PM, Nov 24.**

2 Required Exercises (100 points)

Exercise 1: For the SDD in Figure 1, give annotated parse trees for the following expressions:

1. $(3 + 4) * (5 + 6)n$ [20points]
2. $1 * 2 * 3 * (4 + 5)n$ [20 points]
3. $(9 + 8 * (7 + 6) + 5) * 4n$ [20 points]

	PRODUCTION	SEMANTIC RULES
1)	$L \rightarrow E \mathbf{n}$	$L.val = E.val$
2)	$E \rightarrow E_1 + T$	$E.val = E_1.val + T.val$
3)	$E \rightarrow T$	$E.val = T.val$
4)	$T \rightarrow T_1 * F$	$T.val = T_1.val \times F.val$
5)	$T \rightarrow F$	$T.val = F.val$
6)	$F \rightarrow (E)$	$F.val = E.val$
7)	$F \rightarrow \mathbf{digit}$	$F.val = \mathbf{digit.lexval}$

Figure 1: Syntax-directed definition of a simple desk calculator

Exercise 2: What are all the topological sorts for the dependency graph of Figure 2? One sort mentioned during lecture is 1,2,3,...,9 (slide #17 of Chapter 4). [20 points]

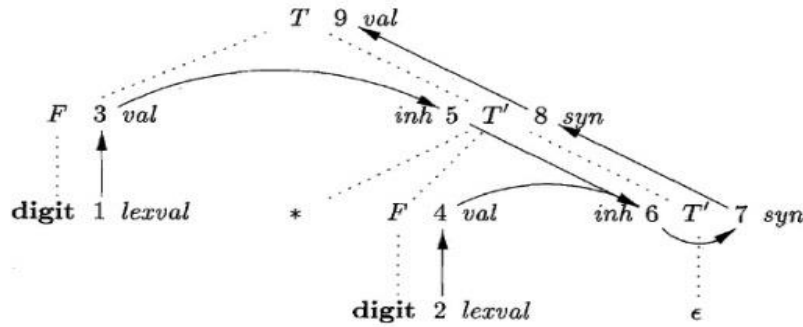


Figure 2: A dependency graph

Exercise 3: Below is a grammar for expressions involving operator + and integer or floating-point operands. Floating-point numbers are distinguished by having a decimal point. Give an SDD to determine the type of each term T and expression E . [20 points]

$$E \rightarrow E + T \mid T$$

$$T \rightarrow num \cdot num \mid num$$

